

Lecture 05: The Character Type

Introduction to Programming with Kotlin

Agenda

1. What is Char?
2. Characters vs strings
3. Codes, ranges, and classification
4. Working with characters in text

The `Char` type

A `Char` holds a **single** Unicode character.

- Literals use single quotes: `'A'`, `'7'`, `'*'`
- Strings use double quotes: `"A"` is a `String` of length 1
- Useful for letters, digits, punctuation, and simple text processing

Declaring characters

```
1 fun main() {  
2     val grade: Char = 'B'  
3     val digit = '7'  
4     val newline = '\n'  
5     val quote = '\''  
6  
7     println(grade)  
8     println("Digit char: $digit")  
9 }
```

Escapes work inside character literals too ('\\n', '\\t', '\\").

Char vs String

	Char	String
Quotes	'A'	"A" / "Hi"
Length	always one char	zero or more
Indexing	—	s[i] yields Char
Typical use	symbols, letters	words, lines, messages

Common mistake

Writing "A" when an API expects Char, or the reverse. The types are different.

Characters from strings

```
1 fun main() {  
2     val word = readln()  
3     if (word.isNotEmpty()) {  
4         val first: Char = word[0]  
5         val last = word[word.length - 1]  
6         println("First: $first, last: $last")  
7     }  
8 }
```

Strings also have `.length` (like `.size` on lists). `word.first()` / `word.last()` are handy when non-empty.

Quiz 1 — Char basics

Quiz

Which literal has type Char?

1. "Z"
2. 'Z'
3. Z
4. char("Z")

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer — Quiz 1

Answer

2. 'z'

Single quotes create a `Char`. Double quotes create a `String`.

Unicode code points (intuition)

Each character has an integer code.

- Digits '0'..'9' are consecutive
- Uppercase 'A'..'Z' are consecutive
- Lowercase 'a'..'z' are consecutive

You can convert: `c.code (Int)` and `Char(code)`.

Codes and arithmetic

```
1 fun main() {  
2     val c = 'A'  
3     println(c.code)           // 65  
4     println(Char(66))        // B  
5  
6     val next = c + 1          // Char arithmetic -> 'B'  
7     println(next)  
8  
9     val digit = '7'  
10    val value = digit - '0'    // 7 as Int  
11    println(value)  
12 }
```

`digit - '0'` is a classic way to turn a digit character into its numeric value 0–9.

Ranges of characters

```
1 fun main() {  
2     val ch = readln()[0]  
3     when (ch) {  
4         in '0'..'9' -> println("Digit")  
5         in 'A'..'Z' -> println("Uppercase")  
6         in 'a'..'z' -> println("Lowercase")  
7         else -> println("Other")  
8     }  
9 }
```

Try this

Count how many letters are in an input line using a for loop over `line` (strings are iterable as characters).

Built-in checks

Kotlin provides helpers on Char:

```
1 fun main() {  
2     val ch = readln()[0]  
3     println(ch.isDigit())  
4     println(ch.isLetter())  
5     println(ch.isLetterOrDigit())  
6     println(ch.isWhitespace())  
7     println(ch.lowercaseChar())  
8     println(ch.uppercaseChar())  
9 }
```

Prefer these for clarity when they match your need.

Quiz 2 — Digit value

Quiz

What is '5' - '0' in Kotlin?

1. '5'
2. 50
3. 5
4. A compile error

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

3. 5

Subtracting two `Char` values yields the distance between their codes as an `Int`. For digit characters, that distance is the digit's value.

Iterate characters of a string

```
1 fun main() {  
2     val line = readln()  
3     var letters = 0  
4     for (ch in line) {  
5         if (ch.isLetter()) {  
6             letters += 1  
7         }  
8     }  
9     println("Letters: $letters")  
10 }
```

Build a string from characters

```
1 fun main() {  
2     val word = readln()  
3     var onlyLetters = ""  
4     for (ch in word) {  
5         if (ch.isLetter()) {  
6             onlyLetters += ch  
7         }  
8     }  
9     println(onlyLetters)  
10 }
```

For heavy concatenation, later you will meet `StringBuilder`; for intro exercises, `+=` is fine.

Caesar-style shift (tiny demo)

```
1 fun main() {  
2     val ch = readln()[0]  
3     if (ch in 'a'..'z') {  
4         val shifted = 'a' + (ch - 'a' + 1) % 26  
5         println(shifted)  
6     } else {  
7         println(ch)  
8     }  
9 }
```

Shifts 'a' → 'b', ..., 'z' → 'a'.

Quiz 3 — Iteration

Quiz

After this code, what is `n`?

```
var n = 0; for (ch in "A1b") if (ch.isLetter()) n++
```

1. 0
2. 1
3. 2
4. 3

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer — Quiz 3

Answer

3. 2

'A' and 'b' are letters; '1' is not.

Summary

- `Char` — one character in single quotes
- Strings are sequences of characters; `s[i]` is a `Char`
- Codes and ranges enable digit/letter checks and simple ciphers
- Prefer `isDigit()`, `isLetter()`, ... when appropriate

Course checkpoint

You can now write Kotlin programs that:

- Read input and use variables / templates
- Compute with integers and floats
- Branch with `if` / `when`
- Loop over lists, ranges, and characters
- Split lines and process text at character level

Next steps in a fuller course: functions, null-safety, classes, and testing.